Windows Azure Storage - Speed and Scale in the Cloud

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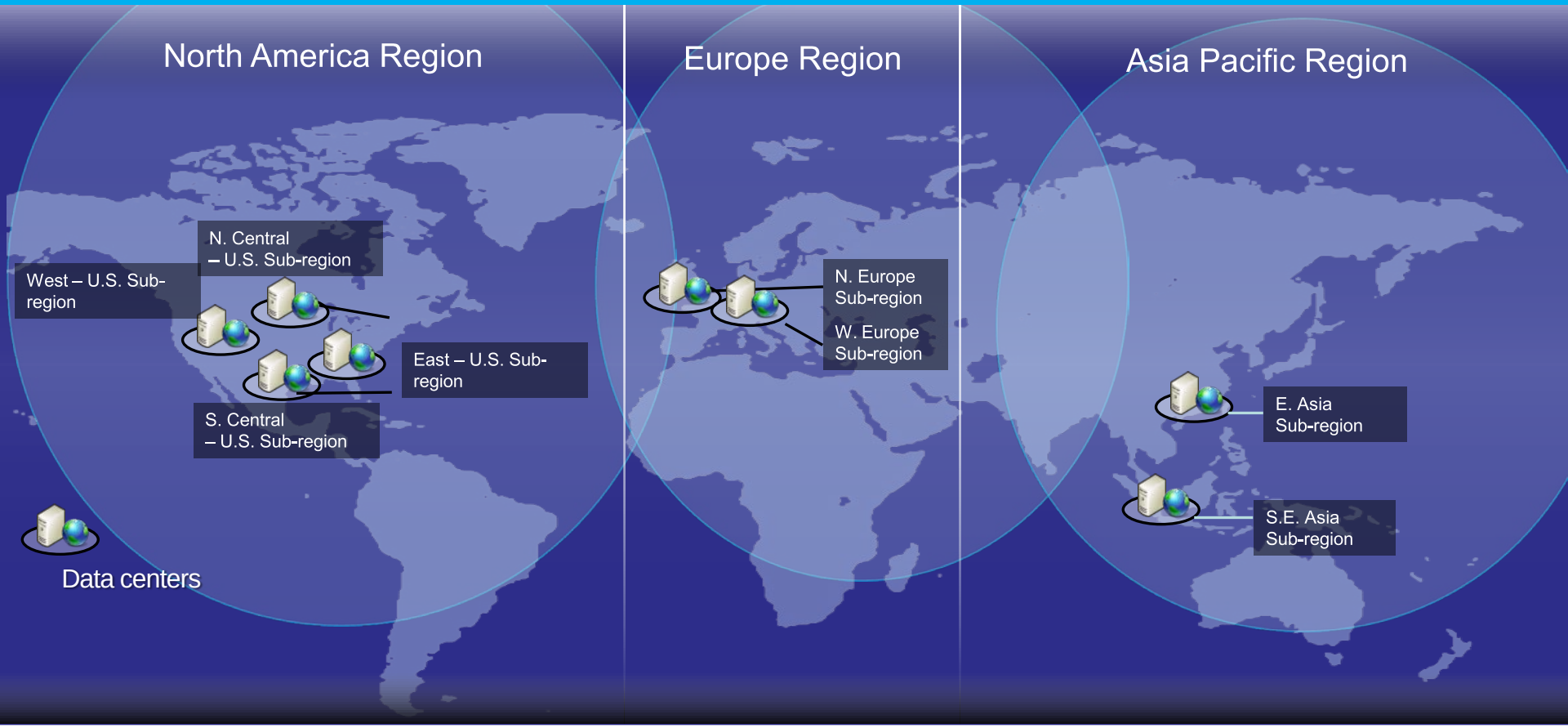
Microsoft

Windows Azure Storage



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Windows Azure Storage

- Abstractions
  - **Blobs** – File system in the cloud
  - **Tables** – Massively scalable NoSQL storage
  - **Queues** – Reliable storage and delivery of messages
  - **Disks** – Durable NTFS volumes for Windows Azure VMs
- Easy client access
  - Easy to use REST APIs and Client Libraries
    - Java, Python, Node.js, .NET, and PHP, and more
    - http://www.windowsazure.com/en-us/documentation/?fb=en-us
  - Existing NTFS APIs for Windows Azure VM Disks

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Characteristics

- **Durable**: Store multiple replicas of your data
  - **Local replication**:
    - Synchronous replication before returning success
  - **Geo replication**:
    - Replicated to data center at least 400+ miles apart
    - Asynchronous replication after returning success to user.
- **Available**: Multiple replicas are placed to provide fault tolerance
- **Scalable**: Automatically partitions data across servers to meet traffic demands
- **Strong consistency:** Default behavior is consistent reads once data is committed

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How we stack up

Nasuni recently released their "State of the Cloud Storage Industry" Report for 2013 in February

- Windows Azure consistently performed better than all other CSPs in the study, which included Amazon S3, Google Cloud Storage, HP Cloud Object Storage, and Rackspace Cloud Files.
- Windows Azure delivered the best Write/Read/Delete speeds across a variety of file sizes, with the fastest response times and fewest errors.
- Windows Azure not only outperformed the competition significantly during the raw performance tests, but it was the only CSP to post zero errors during 100 million reads and writes.

http://www6.nasuni.com/rs/nasuni/images/2013_Nasuni_CSP_Report.pdf

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In the real world

- **Xbox:** Uses Blobs, Tables & Queues for applications like Cloud Game Saves, Halo 4, Music, Kinect data collection etc.
- **SkyDrive:** Uses Blobs to store pictures, documents etc.
- **Bing:** Uses Blobs, Tables and Queues to implement an ingestion engine that consumes Twitter and Facebook public status feeds and provides it to Bing search
- **Skype:** Uses Blobs, Tables and Queues for Skype video messaging
- **HDInsight:** provisions a Hadoop cluster in Azure backed by Blobs

- Clients serving PBs of data
- Trillions of objects
- High Reliability

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Scale

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Scalability Targets

Storage Account scalability targets

- Applies to accounts **created after June 7th 2012**
- **Capacity** – Up to **200 TBs**
- **Transactions** – Up to **20,000** entities/messages/blobs per second

- **Bandwidth for a Geo Redundant storage account**
  - **Ingress -** up to **5 Gbps**
  - **Egress -** up to **10 Gbps**

- **Bandwidth for a Locally Redundant storage account**
  - **Ingress -** up to **10 Gbps**
  - **Egress -** up to **15 Gbps**

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Scalability Targets

Partition level Targets

- **Single Queue** – Account Name + Queue Name
- Up to 2,000 messages per second

- **Single Table Partition** – Account Name + Table Name + PartitionKey
- Up to 2,000 entities per second

- **Single Blob** – Account Name + Container Name + Blob Name
- Up to 60 MB/s

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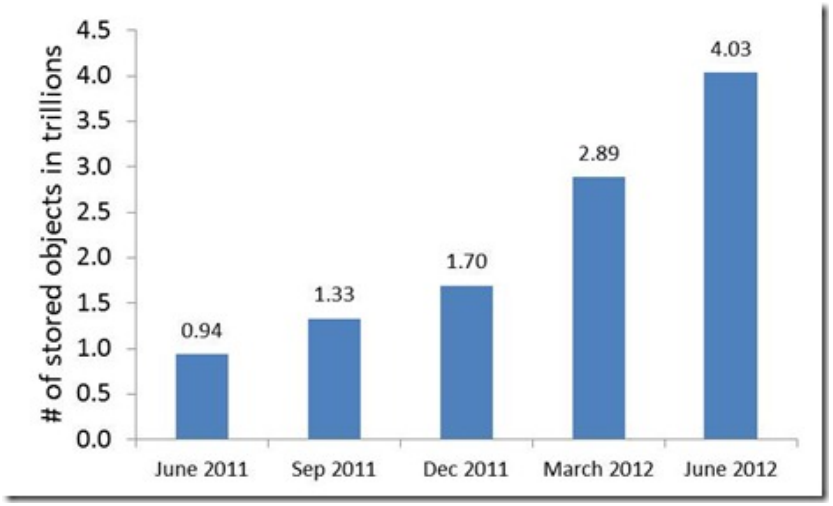
4 Trillion and Counting…

- **4 trillion** objects stored
- **Average** of 270,000 requests/sec
- **Peaks** of 880,000 requests/sec



http://blogs.msdn.com/b/windowsazure/archive/2012/07/18/windows-azure-storage-4-trillion-objects-and-counting.aspx

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Architecture

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Windows Azure Flat Network

- **Flat network storage** design
- **"Quantum 10"** network
- **Non-blocking 10Gbps** based fully meshed network
- Move to **software based Load Balancer**
- **Enables high bandwidth scenarios** such as Windows Azure IaaS disks, HPC, Map Reduce etc.

- http://blogs.msdn.com/b/windowsazurestorage/archive/2012/11/04/windows-azure-s-flat-network-storage-and-2012-scalability-targets.aspx

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Design Goals

- Highly Available with Strong Consistency
  - Provide access to data in face of failures/partitioning
- Durability
  - Replicate data several times within and across data centers
- Scalability
  - Need to scale to zettabytes
  - Provide a global namespace to access data around the world
  - Automatically load balance data to meet peak traffic demands

- Additional details can be found in the following paper:
  - "Windows Azure Storage: A Highly Available Cloud Storage Service with Strong Consistency", *ACM Symposium on Operating System Principals (SOSP)*, Oct. 2011

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23rd ACM SOSP Paper

- Covers in depth the internal design of Windows Azure Storage
  - Approach to CAP theorem in depth
  - Approach to fault tolerance
  - Scalability

http://blogs.msdn.com/b/windowsazurestorage/archive/2011/11/20/windows-azure-storage-a-highly-available-cloud-storage-service-with-strong-consistency.aspx

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Performance

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General Client Best Practices

- Co-locate with storage
- Ensure you have enough simultaneous connections (i.e. node.js, .NET, etc.).
- Disable Nagle for small messages (< 1400 b)
- Disable Expect 100-Continue
- Use Http Keep-Alive
- Use HTTPS for SAS / Sensitive information

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General Server Best Practices

- Locate Storage accounts close to users
  - Co-locate compute in same DC (intra-DC bandwidth is free)
  - For wide spread client apps consider using more than one storage account in different physical DCs
- Understand Account Scalability targets
  - Account Scalability targets are different depending on Geo-Replication
    - Geo Redundant: 5 Gbps ingress / 10 Gbps egress
    - Locally Redundant : 10 Gbps ingress / 15 Gbps egress
- Enable Logging & Metrics on each storage service
  - Can be done via REST, Client API, or Portal
  - Enables clients to self diagnose issues
  - Data can be automatically GC'd according retention policy

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General Server Best Practices

- Optimize what you send & receive
  - Blobs: Range reads, Metadata, Head Requests
  - Tables: Upsert, Merge, Projection, Point Queries
  - Queues: Update Message, Batch size
- Distribute load over many partitions
  - Note: Table Entity Group Transaction (batch) only applies to entities in the same partition
- Control Parallelism at the application layer
- Blend Traffic Types
  - In general Blob workloads tend to be IO bound as Table workloads tend to be CPU limited. By blending traffic in roles you can make the most of your compute resources

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Blobs

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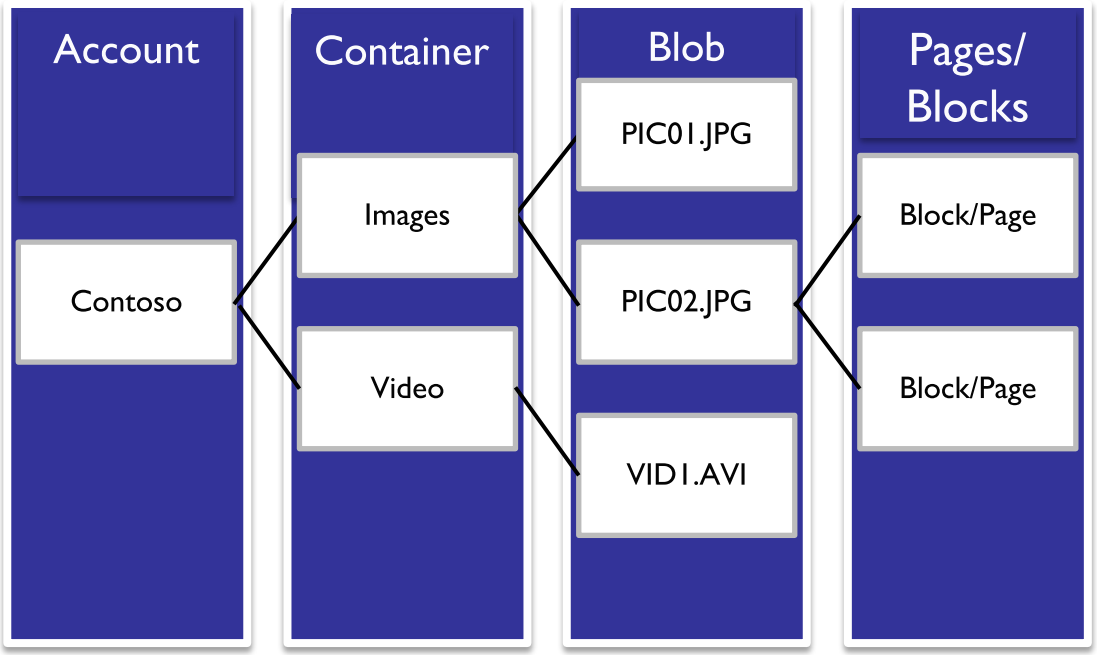
Windows Azure Blobs

http://<account>.**blob**.core.windows.net/<container>/<blobname>



BLOB Storage is the simplest way to store large amounts of unstructured text or binary data such as video, audio and images with the fastest
read performance.

Highly scalable, durable, available file system.

Blobs can be exposed publically over http.

Can securely lock down permissions to blobs.

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Blob Containers

- Number of Blob Containers
  - Can have has many Blob Containers that will fit within the storage account limit
- Blob Container
  - A container holds a set of blobs
  - Set access policies at the container level
    - Private or Public accessible or Store Shared Access Identifiers
  - Associate Metadata with Container
    - Metadata are <name, value> pairs
    - Up to 8KB per container
  - List the blobs in a container

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Types of Blobs

- Block Blob
  - Targeted at streaming workloads and general purpose files
  - Each blob consists of a sequence of blocks
    - Each block is identified by a Block ID
  - Size limit 200,000 MB per blob
- Page Blob
  - Targeted at random write workloads
  - Each blob consists of an array of pages
    - Each page is identified by its offset from the start of the blob
  - Size limit 1TB per blob
  - Must be 512 byte aligned

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Block Blob Details

- Block can be up to 4MB each
  - Each block can be variable size
  - Each block has a 64 byte ID
- Block operation
  - PutBlock - Puts an uncommitted block for the blob
- Block List Operations
  - PutBlockList - Provide the list of blocks to comprise the readable version of the blob
    - Can use blocks from uncommitted or committed list for updates
  - GetBlockList - Returns the list of blocks, committed or uncommitted
    - Block ID and Size of Block is returned for each block

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Page Blob Details

- Page Blob is created with a Max Blob Size
  - Can change the max size of the blob at anytime.
- Address space is broken up into 512 byte pages for updates
- Page update operations – must be page aligned
  - PutPage - limited to 4MB
    - Overwrite range of pages starting at the specified offset
  - ClearPage - can specify up to max size of blob
    - Clear range of pages at the offset
- Reading a Page Blob
  - GetBlob – can read from any byte offset for any valid range
- What parts of the Page Blob have stored pages in them
  - GetPageRanges - Get valid page ranges in the blob
- Only charged for pages with data stored in them

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Blobs – Put Throughput

- A single XL VM can achieve 1 Gbps to the blob store. A single partition can provide 60 MB/s
- Upload Multiple blobs concurrently using minimal blob parallelism
  - Distributes load over many partitionsBlobs under 64 MB can be uploaded as a single put, otherwise multiple block requests & a put block list are required
  - Avoids "long tail" problem – 11 MB blob with1MB blocks, parallelism =10

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Blobs – Get Throughput

- In general larger read sizes are better for throughput
  - Less overhead for connection establishment / Auth
  - Allows TCP window to grow
- Example: 256MB Block Blob with 4 MB blocks
  - Single Get averaged 84.46 MB/s
  - 4MB Read averaged 26.9 MB/s
  - 64 KB Read averaged 8.69 MB/s
- Note: A BlockBlob can contain 50,000 Blocks
  - 4 MB = Max of 195.31GB
  - 2 MB = Max of 97.65GB
  - …
  - 128KB = Max of 6.1 GB

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Simple Client API example (java)

```
// Retrieve storage account from connection-string CloudStorageAccount storageAccount =
CloudStorageAccount.parse([CONNECTION STRING]);

// Create the blob client
CloudBlobClient blobClient = storageAccount.createCloudBlobClient();

// Create a container
CloudBlobContainer container = blobClient.getContainerReference("mycontainer");
container.create();

// Upload a blob with contents from a local file
CloudBlockBlob blob = container.getBlockBlobReference("myimage.jpg");
File source = new File("c:\\myimages\\myimage.jpg");
blob.upload(new FileInputStream(source), source.length());
```

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Tables

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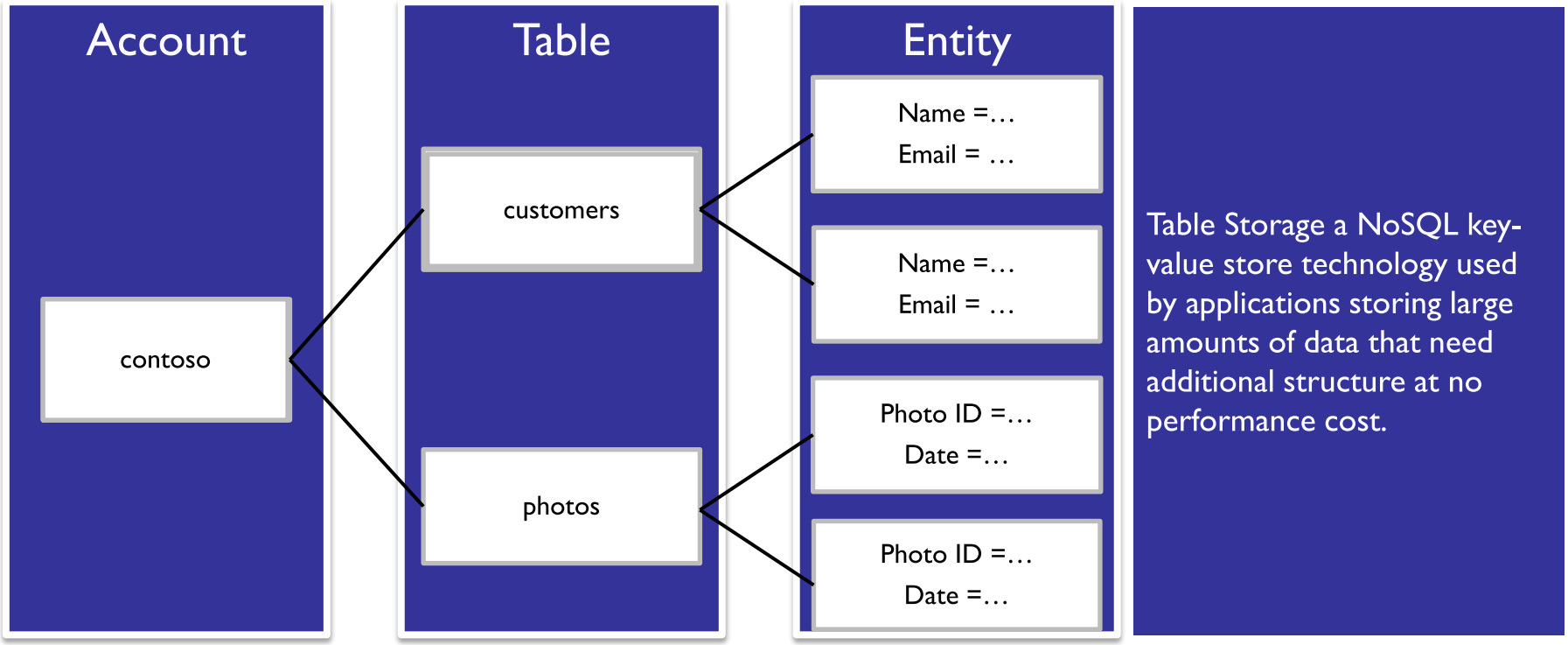
Windows Azure Tables

http://<account>.**table**.core.windows.net/



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Windows Azure Table

- Flexible schema
  - Tables – No schema associated
  - Store different entity types in same table
- Single Index
  - (AccountName, TableName, PartitionKey, RowKey) => Clustered Index
  - PartitionKey – Mandatory property which defines partitioning
  - RowKey – Mandatory property which defines uniqueness within a given partition

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Windows Azure Tables

- Optimistic concurrency
  - Timestamp – A read only property maintained by server
- Query
  - Single entity lookup
  - Scans: Range based query
  - Performance depends on selectivity
- Atomic Transactions
  - Entities with same partitioning key can be in a batch request
- Exposed via RESTful APIs
  - OData protocol
  - Java, Node.js, PHP, .NET client libraries

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Window Azure Tables

- All Entities define 3 reserved properties
  - PartitionKey
  - RowKey
  - Timestamp
- Entities can contain 255 properties
- Properties can be up to 64 KB each
- A single entity can be up to 1 MB
- A single atomic batch operation can be up to 4 MB

http://blogs.msdn.com/b/windowsazurestorage/archive/2010/11/06/how-to-get-most-out-of-windows-azure-tables.aspx

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Table Performance Best Practice

- Key selection is single most important performance decision you will make!
  - Entity Group Transaction
  - Scalability
  - Efficient queries
- Send and receive the smallest amount of data necessary
  - Use Merge / Upsert when appropriate
  - Server side projection

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Query Design & Selectivity

- Understand Continuation tokens
  - Returned at 1000 entities
  - 4 MB payload
  - Server boundaries
- Table Scans are expensive – avoid them for latency sensitive scenarios
- Performance of queries
  - Point Query (PK, RK) - Fastest
  - Row Key Range Query(PK, A < RK< D)
  - Row Key Range Query Open ended (PK, A < RK)
  - Multi Partition Query (A <PK < D, RK …)
  - Query on non-indexed column (Timestamp, user column, etc.)

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Tables - Throughput

- Batch Size (Entity Group Transaction)
  - Use Batch to perform atomic actions on a given partition (all entities in the same table with the same PK)
  - Up to 100 entities
  - Up to 4 MB in size
  - Dramatically improves throughput and per entity latencies
- Insert operations return the entity back to the client Merge, Replace, and Upserts do not.
  - Less bytes to read => less bytes to parse => lower NIC / CPU => Higher throughput

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Tables - Latency

- Disable Nagle for small entities (1400 bytes)
- Query Selectivity
  - Smaller take count / page sizes for latency sensitive applications
- Projection: Only take what you need
- Merge: Only send what changed

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Queues

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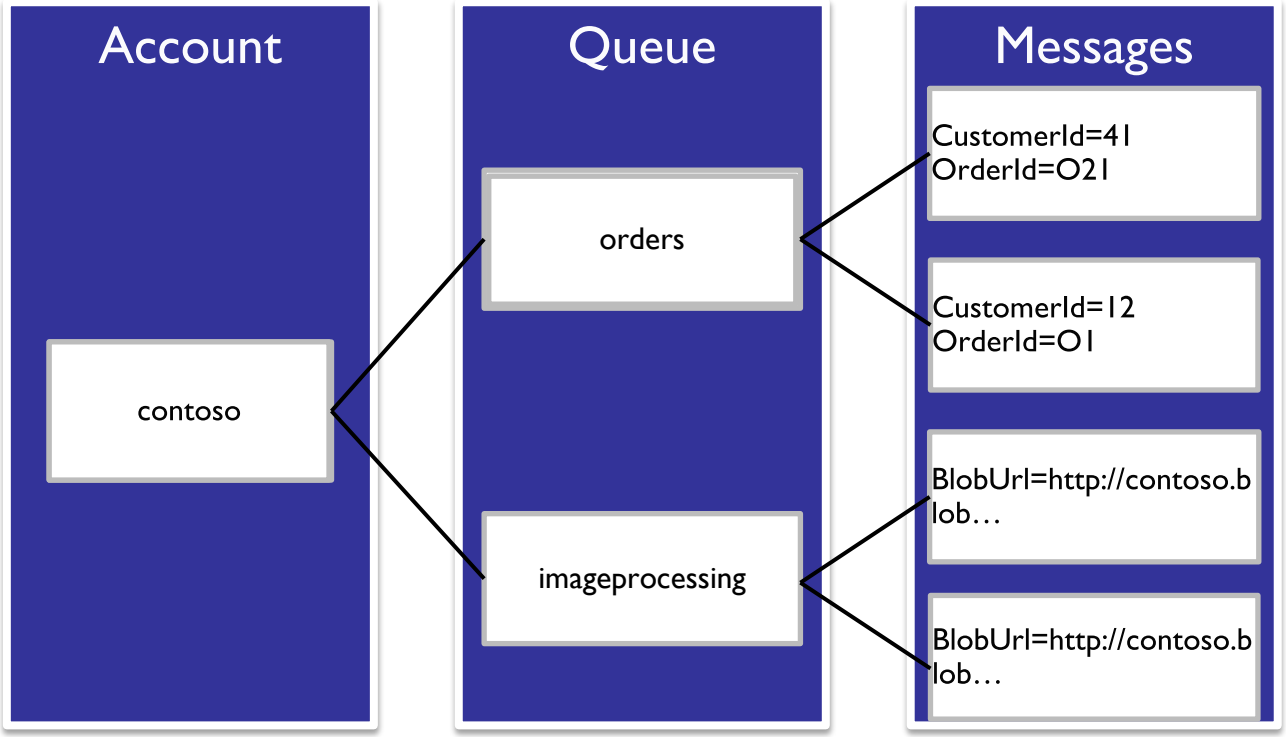
Windows Azure Queues

http://<account>.**queue**.core.windows.net/



Queues offer a guaranteed message delivery mechanism that can be used to distribute work across roles. Utilizing a two-phase delete process ensures that even if a role which retrieves a message goes down the message will become visible again and subsequently processed by another worker.

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Windows Azure Queues

- Create, Delete, SetAcl Queues
- Enqueue Messages – up to 64KB messages
- Lease mechanism for processing
  - Get Message(s) – provide lease time
  - Delete Message
- Metadata
  - Message count – Enables auto scaling
  - Dequeue count – Detect poison messages
- Sharing Scenarios
  - Private access or Shared Access Signatures (Signed Url)

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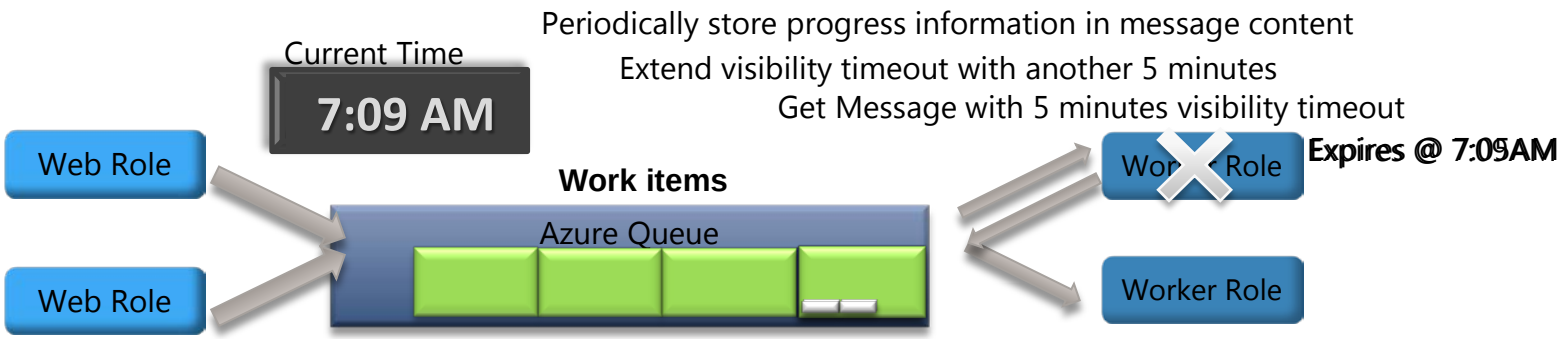
Message Processing



Retrieve progress from queue message and resume

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Queues Performance Best Practice

- Nagle is important as many messages may be small
- Design application to update messages rather than deleting and inserting a new one
- A queue is a single partition, for scale use multiple queues
- Use Message de-queue count to detect malformed data or "poison" messages

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Queues - Throughput

- Use batching GetMesssages when appropriate
- Tune Initial Invisibility Time & Visibility Timeout
  - Shorter Visibility Timeout will require a more responsive service, but can ensure lower latencies, longer visibility timeouts
- Tune Message Count for Peek & Get
  - Consider failure modes, if a worker gets 32 messages but fails immediately 31 messages must also wait to become visible again
- Use Multiple Queues

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Queues - Latency

- Disable Nagle
- Tune Initial Invisibility Time & Visibility Timeout
  - Shorter Visibility Timeout will require a more responsive service, but can ensure lower latencies, longer visibility timeouts can allow larger amounts of processing to be done before check pointing
- Use smaller Message Count for Peek & Get
- Http keep-alive when doing many small requests

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Summary

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Future Directions

- CORS (Cross origin resource sharing)
- Continuous performance improvements
- Expanding support for additional client platforms
- Geo-Replication for Queues
- Read from secondary DC
- Expanded Reach – New Region build outs
  - http://blogs.technet.com/b/microsoft_blog/archive/2013/05/22/microsoft-announces-major-expansion-of-windows-azure-services-in-asia.aspx
- And more …

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Resources

- Storage team blogs @ http://blogs.msdn.com/b/windowsazurestorage/
- Getting Started @ https://www.windowsazure.com/en-us/develop/overview/
- Pricing information @ https://www.windowsazure.com/en-us/pricing/details/storage/
- Open Source Client Libraries @ www.github.com/WindowsAzure

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Questions?

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